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सिपेट: आई पी टी-कोच्ची /परीक्षण/२०२३-२४ **दिनांक: १७.०२.२०२४**
CIPET: IPT-Kochi /Testing/2023-24/3278 **Date: 17.02.2024**

सेवा में / To

M/s. Alfa Paper Products
Athanikkal, Vailathur, Ponmundam P O
Malappuram Dist – 676 106

प्रिय महोदय / Dear Sir,

विषय: नमूनों की परीक्षण - संदर्भ में।
Sub.: Testing of Samples
Ref.: Letter dt 26.04.2023

हम इसके साथ परीक्षण के लिए प्रस्तुत नमूने से संबंधित टेस्ट रिपोर्ट 231073 दिनांकित 17.02.2024 संलग्न कर रहे हैं।

We are enclosing herewith Test Report No. 231073 dated 17.02.2024 pertaining to the samples submitted for testing.

कृपया संलग्न कस्टमर फीडबैक फॉर्म भरकर वापस भेजने का कष्ट करें।
Please find enclosed herewith the feedback form. Kindly fill and send it back to us.

धन्यवाद तथा सबसे अच्छी सेवा देने का आश्वासन के साथ।
Thanking you and assuring you our best services.

सादर, / Yours sincerely,

संयुक्त निदेशक और प्रमुख
Joint Director & Head

संलग्न यथोक्त / Encl. as above.

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परीक्षण रिपोर्ट
TEST REPORT

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क्र.सं. / Sl.No. 11200

ANALYSIS REPORT

Issued to :

M/s. Alfa Paper Products
Athanikkal, Vailathur, Ponmundam P O
Malappuram Dist – 676 106

Test Report No : 231073
Date: 17.02.2024

Customer Ref. No. & date : Letter dt 26.04.2023
Work order Ref.No. : 072/23-24
As per Standard: : As per part C

PART A : PARTICULARS OF SAMPLE SUBMITTED

a)Name of the sample	:	Paper board with Compostable polymer coating for manufacturing of cups and plates as stated by the party
b)Grade / Variety / type / Size / Class etc.	:	Nil
c)Code No.	:	Nil
d)Quantity (pcs/mtr/gm/nos)	:	2 Kg
e)Mode of Packing (Sealed cartoon/polypouch/container or not)	:	Sealed carton
f)Date of receipt of sample	:	26.04.2023
g)Date of Performance of test	:	31.05.2023 -08.02.2024
h)Any other information	:	Interim Report No. 23695 dt. 21.11.2023
i)Unique Sample ID	:	2304113

PART B: SUPPLEMENTARY INFORMATION

a) Reference to sampling procedure	:	Drawn & Supplied by the party
b) Supporting documents for measurements taken and results derived like graphs, tables, sketches and / or Photographs as appropriate to test report, if any (to be attached)	:	As per part -C
c) Deviation from the test methods as prescribed in relevant ASTM/ISO/BIS / Work instructions, if any	:	---

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Manager (Technical)
Authorized Signatory

Dr. S. Anbudayanidhi
Manager (Technical)
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TEST REPORT

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क्र.सं. / Sl.No. 11200

ANALYSIS REPORT

PART C: TEST RESULTS

Report No.:231073
Date: 17.02.2024

Sl. No	Name of the test	Test Method/ Standard	Unit	Results Obtained	Specified Requirement as per ISO : 17088-2021 and ASTM D 6868
Sample details: Paper board with Compostable polymer coating for manufacturing of cups and plates as stated by the party					
1.	Material Identification	FTIR/DSC	--	Base Material : Paper (Cellulose) Coating: Poly (Butadiene - co- grafted Acrylonitrile)	--
2.	Disintegration (Dry mass remains in 2mm sieve after 84 days)	ISO 20200:2015	%	8.90	Not more than 10% of its original dry mass
3.	Ultimate aerobic biodegradation (with reference to 100% degradation of positive reference)	ISO 14855-1:2012	%	91.31 (at the end of 154 days)	> 90% (At the end of the test period not more than 180 days)
4.	Plant Growth study **Monocotyledon(Paddy)** % Seed emergence **Dicotyledon(Tomato)** % Seed emergence	OECD : 208 Guidelines	% %	91 91	> 90% of those from the corresponding blank compost

Note: The detailed observation on biodegradability test is enclosed as Annexure. I

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ANALYSIS REPORT

PART C: TEST RESULTS

Report No.:231073
Date: 17.02.2024

Sl. No	Name of the test	Test Method/ Standard	Unit	Results Obtained	Specified Requirement as per ISO : 17088-2021 and ASTM D 6868
5.	Acute Ecotoxic Effects to earthworm				
a.	Survival of adult earthworm at the end of 7 days	ISO 11268-1:2012	%	90	> 90% of those from the corresponding blank compost
b.	Survival of adult earthworm at the end of 14 days		%	91.56	
c.	Biomass at the end of 14 days		%	91.08	
6	Chronic Ecotoxic Effects to earthworm				
a.	Survival of adults earthworm at the end of 28 days	ISO 11268-2:2023	%	96.38	> 90% of those from the corresponding blank compost
b.	Offspring at the end of 56 days		%	92.69	
c.	Biomass at the end of 56 days		%	91.79	

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Continuation Sheet

ANALYSIS REPORT

Report No.: 231073
Date: 17.02.2024

PART C: TEST RESULTS

Sl. No.	Property	Test method / Standard	Unit	Results obtained	Specified Requirements (Max.) as per IS 17899 T:2022
7.	Heavy Metal Analysis (on dry mass basis)				
	Arsenic (As)	IS 3025 (Part 37):2022	mg / kg	0.0046	10
	Copper (Cu)	IS 3025 (Part 42):1992		1.1553	300
	Nickel (Ni)	IS 3025 (Part 54):2003		0.7520	50
	Zinc (Zn)	IS 3025 (Part 49):1994		1.2990	1000
	Chromium (Cr)	IS 3025 (Part 52):2003		0.6919	50
	Mercury (Hg)	IS 3025 (Part 48):1994		0.0157	0.15
	Cadmium(Cd)	IS 3025 (Part 41):2023		0.9388	5
	Lead (Pb)	IS 3025 (Part 47):1994		1.4327	100

PART D: REMARKS: NIL

Note
1. This Test Report / Certificate is issued only for the samples submitted to the laboratory.
2. The results stated above related only to the items tested.
3. The quality of the subsequent production lot has to be ensured by the purchaser.
4. This Test Report shall not be reproduced except in full without the written approval of the laboratory.
5. Any anomaly/discrepancy in this report should be brought to the notice of the laboratory within 30 days from the date of issue.
6. Subcontracted Tests (if any): S.No.1

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ANNEXURE-I

TR. NO.:231073 **ANALYSIS RESULT** **Date: 17.02.2024**

OBSERVATION FOR BIODEGRADABILITY TEST AS PER ISO 17088:2021

Name of the Customer :

M/s. Alfa Paper Products
Athanikkal, Vailathur, Ponmundam P O
Malappuram Dist – 676 106.

1.Sample Details: Paper board with Compostable polymer coating for manufacturing of cups and plates as stated by the party.

2. Material Identification by FTIR & DSC: Base Material : Paper (Cellulose)
Coating:Poly (Butadiene - co- grafted Acrylonitrile)

3. Observations:

a. Conditions of reaction Mixture

Origin of Compost	: Livestock excrement, municipal and vegetable waste
Reaction Temperature	: 58°C (±2°C)
Dry Solid (%)	: 54.58 %
Volatile content (%)	: 30.69%
CO_2 evolved during 1st 10 days in blank vessels	: 54.89 mg/g of volatile solids of compost
Test Duration (Days)	:154 Days
Reference material	: Cellulose
Volume of reaction Vessel	: 3000ml

DR. K. P. BHUVANA
MANAGER (TECHNICAL)
AUTHORIZED SIGNATORY

DR. S. ANBUDAYANIDHI
MANAGER (TECHNICAL)
AUTHORIZED SIGNATORY

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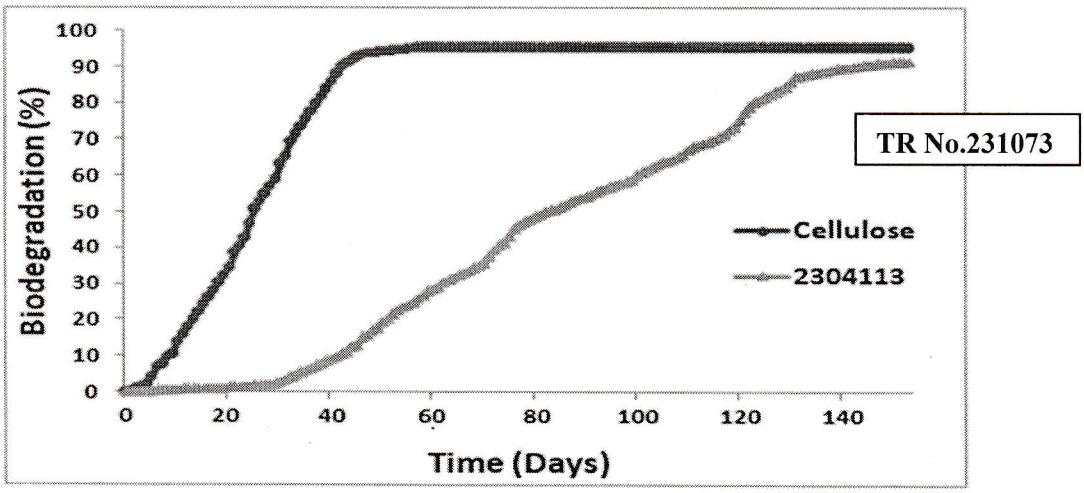
b. pH of test medium

Sl. No	Composting Vessel (Material with test medium)	pH (Before)	pH (After)
1	Sample 1	7.4	7.5
2	Sample 2	7.3	7.5
3	Sample 3	7.3	7.4
4	Blank	7.1	7.3
5	Blank	7.1	7.2
6	Blank	7.3	7.4
7	Positive 1	7.7	7.7
8	Positive 2	7.6	7.7
9	Positive 3	7.5	7.6
10	Negative	7.5	7.6
11	Negative	7.6	7.7
12	Negative	7.5	7.6

4. Result: Percentage biodegradation relative to positive reference

Mean(%) : 91.31%

The reference material- cellulose(%) : ~100%



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Continuation Sheet

TR. NO.:231073 **ANALYSIS RESULT** **Date: 17.02.2024**

5. Visual observation of Sample

Description	Week 1	Week 7	Week 13	Week 18	Week 22
Structure	Cut Pieces of paper cup sample	Cut Pieces of paper cup sample	Disintegrated pieces of paper cup	Disintegrated pieces of paper cup	--
Moisture	Adequate moisture level	Adequate moisture level	Adequate moisture level	Adequate moisture level	Adequate moisture level
Colour	Off-white	Faded White	Faded White	Faded White	--
Fungal Development	Nil	Nil	Nil	Nil	Nil
Smell	Organic/ Dirt Like	Organic/ Dirt Like	Organic/ Dirt Like	Organic/ Dirt Like	Organic/ Dirt Like

6. Visual observation of compost

Description	Week 1	Week 7	Week 13	Week 18	Week 22
Structure	Fine Particles	Fine Particles	Fine Particles	Fine Particles	Fine Particles
Moisture	Adequate moisture level	Adequate moisture level	Adequate moisture level	Adequate moisture level	Adequate moisture level
Colour	Dark Brown	Dark Brown	Dark Brown	Dark Brown	Dark Brown
Fungal Development	Nil	Nil	Nil	Nil	Nil
Smell	Organic/ Dirt Like	Organic/ Dirt Like	Organic/ Dirt Like	Organic/ Dirt Like	Organic/ Dirt Like

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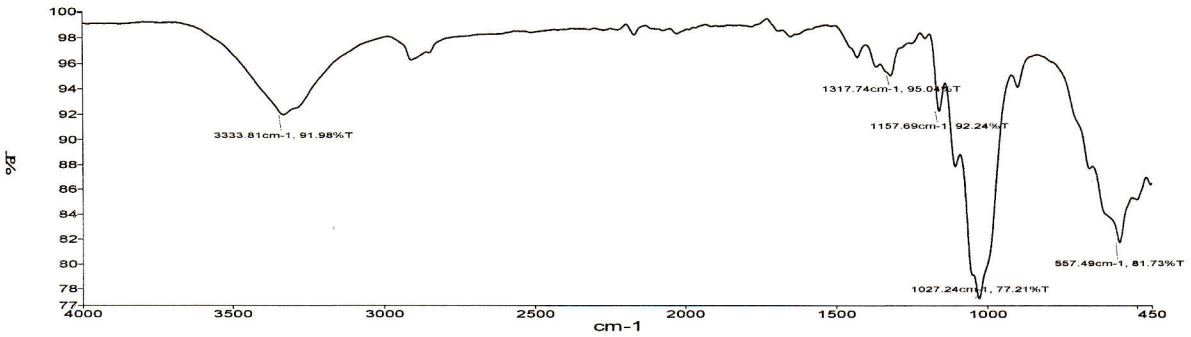
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TR. NO.:231073 **ANALYSIS RESULT** **Date: 17.02.2024**

Sample Details: Paper board with Compostable polymer coating for manufacturing of cups and plates as stated by the party.

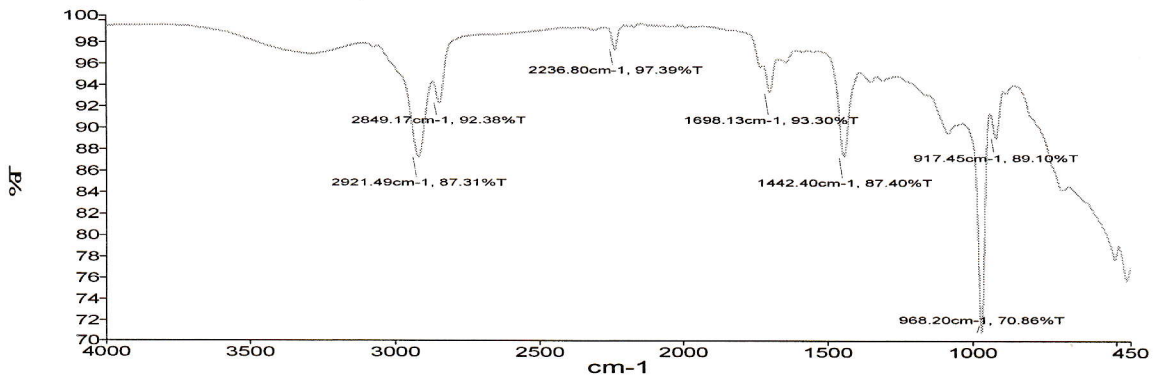
7. FTIR Analysis

FTIR Spectrum of outer layer



Wave number(cm^{-1})	Nature of Bond
3333.81	O-H stretching vibration
1317.74	O-H bending vibration
1157.69	C-O stretching vibration
1027.24	C-O stretching vibration

FTIR Spectrum of Coating



Wave number(cm^{-1})	Nature of Bond
2921.49	C-H stretching vibration
2236.80	C≡N stretching vibration
1698.13	C=O stretching vibration
968.20	C=C bending vibration

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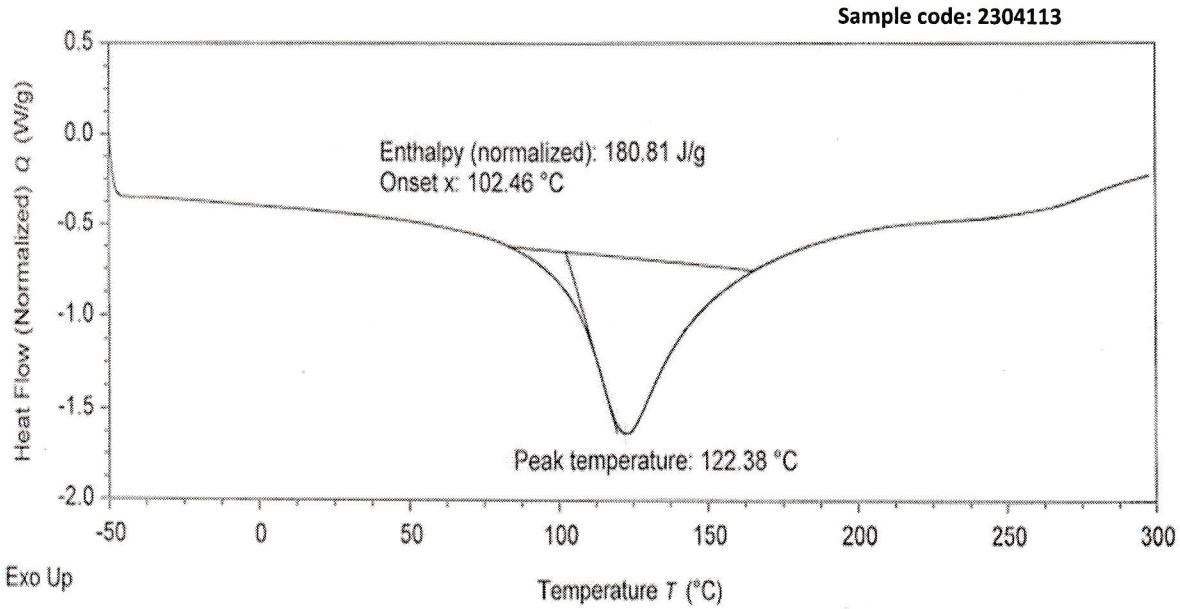
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Continuation Sheet

TR. NO.:231073 **ANALYSIS RESULT** **Date: 17.02.2024**

8. DSC Analysis



Comment: The above DSC & FTIR analysis indicates the above sample is cellulose coated with Poly (Butadiene - co- grafted Acrylonitrile)

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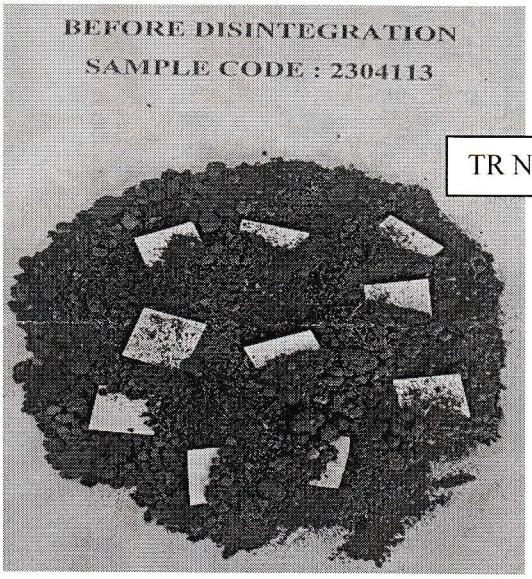
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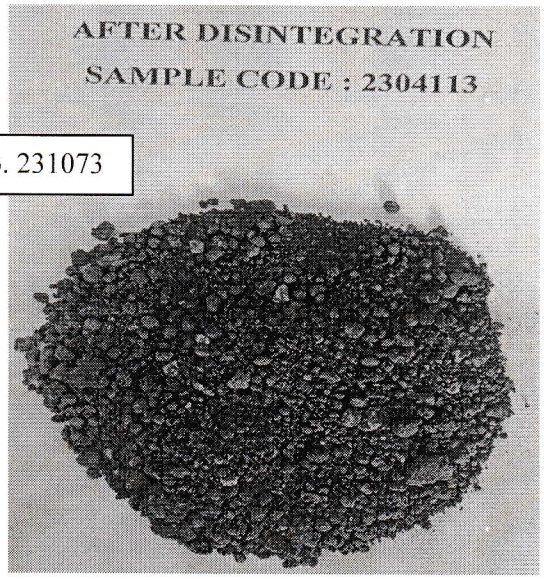
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TR. NO.:231073 **ANALYSIS RESULT** **Date: 17.02.2024**

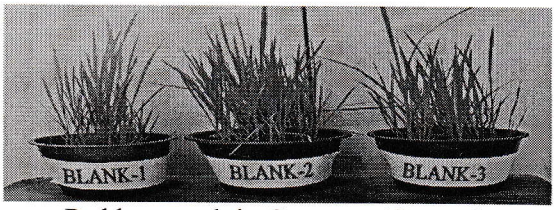
9. DISINTEGRATION- AFTER 12 WEEKS



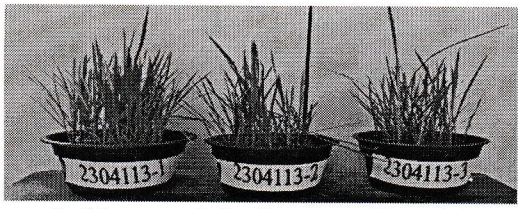


The disintegration of the supplied sample by passing through 2 mm sieve after 12 week in composting condition as per ISO 17088-2021 was found not more than 10% of original dry mass remain.

10. SEEDGERMINATION AND PLANT GROWTH STUDY



Paddy growth in Compost (Control)



Paddy growth in Compost (Sample)



Tomato growth in Compost (Control)



Tomato growth in Compost (Sample)

The percentage of seed germination was found to be greater than 90% for both control and sample.

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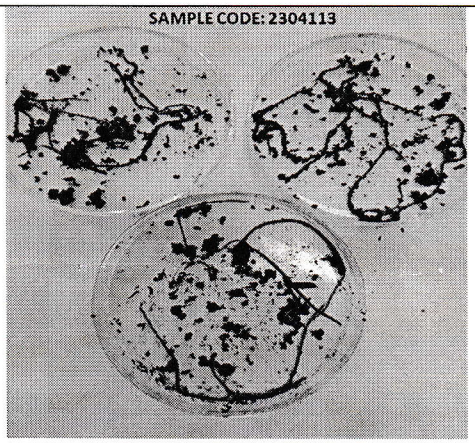
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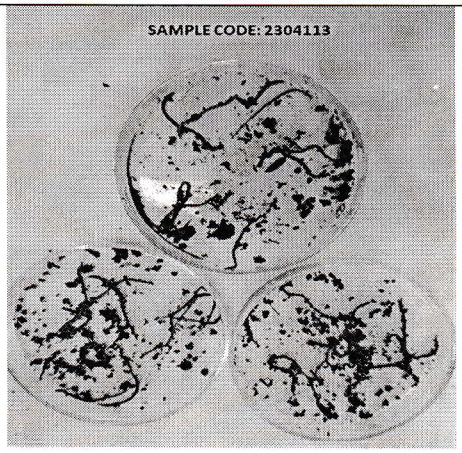
TR. NO.:231073 **ANALYSIS RESULT** **Date: 17.02.2024**

11. Acute & Chronic Ecotoxicity effects to Earthworm

<table>
<tr>
<td> 

Photograph of Live earthworm in the sample compost at the end of 7 days</td>
<td> 

Photograph of Live earthworm in the sample compost at the end of 14 days</td>
</tr>
<tr>
<td colspan="2">The surviving adult earthworms grown in the sample compost exposed to the test material after an incubation period of 14 days is more than 90 % of those from the corresponding blank compost not exposed to any material.</td>
</tr>
<tr>
<td> 

Photograph of Live earthworms in the sample compost at the end of 28 days.</td>
<td> 

Photograph of Live earthworm in the sample compost at the end of 56 days.</td>
</tr>
<tr>
<td colspan="2">The surviving adult earthworms grown in the sample compost exposed to the test material after an incubation period of 28 days and the counted number of offspring after an incubation period of 56 days is more than 90 % of those from the corresponding blank compost.</td>
</tr>
</table>

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